The logo's left and right ends evoke the horizontally opposed pistons of the boxer engine flat design.

> As long as I can remember, I've always wanted to make games. Somehow I ended up doing it, and a good way for me to enjoy this journey is to build my own game engine and share it with you, hoping it will help you too.

Boxer

"The Boxer project draws its name from the boxer engine architecture found in cars, symbolizing efficiency in a compact form."

A minimalist, modern, and lightweight 2D game engine written in C, designed so you can easily build your game around it, favoring direct inclusion rather than linking as a static or shared library.

Prerequisites

Install the following development dependencies (via your package manager or from source):

- GNUMake.
- C compiler.
- SDL3 (>= 3.4.2).
- SDL3_image (>= 3.0.0).
- SDL3_mixer (>= 3.0.0).
- SDL3_ttf (>= 3.0.0).

Build and Run

Clone the repo:

```
git clone git@github.com:Livy-s-Quest/boxer.git
cd boxer
```

Debug build:

```
make DEBUG=1
```

Release build:

```
make
```

Run the example game:

```
./my_game
```

let's make it roar!

Project Structure

The project is organized as follows:

```
data/  # Game assets such as textures, audio, etc.
boxer/ # Core engine source code
game/  # Your game source code
test/  # Unit tests for the engine (take a look for examples of how to use the API)
```

Project Filesystem

Under its hood (or head cover since it's an engine), Boxer uses PhysFS for virtual filesystem management. By default, the following folders are mounted:

- `data.zip` (containing all your game data) is mounted as `data/` **(read-only)**. This is where the game should load assets from, such as textures, audio, etc.
- `the system preferred path` is mounted as `/` **(read/write)**. This is where the game should save user data, such as save files, settings, etc.

This means that files are read from both `data.zip` and `the system preferred path`, but are only written to `the system preferred path`. This allows the game to have a **read-only** assets folder and a separate **read/write** user data folder.

Depending on the operating system, the system preferred path is as follows:

- **Linux:** `~/.local/share/<config->name>/`
- **Windows:** `C:\Users\<username>\AppData\Roaming\<config->name>\`
- **macOS:** `~/Library/Application Support/<config->name>/`

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